

StaleLine: A Passive Maker on polymarket.com Football (HYPOTHESIS_v4)

Our live hypothesis: high risk, high reward, untested so far

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Abstract

HYPOTHESIS_v4 is our live hypothesis: high risk, high reward, and not yet tested.

A passive maker on polymarket.com football joins the best bid and ask (Arm A always; Arm B only while the spread is at least 3 cents), frozen at 07:56 ET, Oct 4, 2026 (94aa1a0) before any test data. Its origin is one simulated day on recorded top of book (Oct 3, 88 games, maker fee 0): +\$46.53 at +60 s, 1.09 cents per contract [0.17, 2.12], excluding the five largest games the estimate stays positive at +0.15 cents per contract [-0.48, 0.77] (condition (2) of v4's own decision rule), and 57.6% of the 59 games with fills were positive (38.6% of all 88). Under our reading rule (games with fills) all four conditions pass: the +60 s interval excludes zero, so we treat it as a real lead that needs more testing on live games. The confirmatory test runs once on Oct 8 to 12 games. v4 grew out of an earlier pre-registered Kalshi study (the v3 strategies, not v4) that did not survive its single holdout test, after which 566 disclosed trials found no other edge. Taking liquidity does not pay after costs, so v4 crosses into largely uncharted territory: is a patient maker paid for supplying liquidity to recreational sports flow? The reward is an edge that grows with the spread; the risk is one day of evidence.

1 The live hypothesis: HYPOTHESIS_v4 (high risk, high reward)

Rule (frozen, 94aa1a0). On polymarket.com (and, for the confirmatory test, Polymarket US through its authenticated order-book websocket), quote both sides at the best bid and ask, joining and never improving, 10 contracts, inventory cap ± 50 , no Kalshi signal. Arm A always quotes; Arm B quotes only while the spread is at least 3 cents. Fills only on strict trade-through; published maker fees; kickoff window cut. Primary metric: P&L per contract marked to mid at +60 s, game-bootstrap 95% CI (seed 20261011). **Economic logic:** our earlier lead-lag test (Section 2) found no quote-level lag between Kalshi and polymarket.com, so a resting quote is rarely stale when hit; a maker is paid the spread for providing liquidity to recreational flow, and the edge persists if spread capture exceeds adverse selection [2].

Origin evidence (descriptive, one day, not pre-registered, simulated on recorded top of book, maker fee 0, queue position unknown). The plain maker on Oct 3 (88 games, 59 with fills, 428 fills) made +\$46.53 at +60 s and +\$114.03 held to settlement; 1.09 cents per contract [0.17, 2.12] at +60 s and 2.66 cents [-0.62, 6.06] at settlement. It captured \$63.18 of spread (1.63 cents per contract) and lost \$15.65 (0.40 cents) to adverse selection. It loses at 1 cent spreads (-0.62 cents) and wins at 3 cents or more (+2.73), hence Arm B. Per-game Sharpe (mean over standard deviation of game P&L, not annualized) is 0.22 [0.06, 0.35] at +60 s and 0.16 [-0.04, 0.35] at settlement; one day supports no annualized figure. Five games carry \$41.75 of the \$46.53 (Figure 1); excluding the five largest games, the estimate stays positive at +0.15 c per contract [-0.48, 0.77]; this is condition (2) of v4's own decision rule. 57.6% of the 59 games with fills were positive (38.6% of all 88). Our ruling: under the reading rule applied to the 59 games with fills, all four conditions pass (interval above zero, positive without the top five games, a majority of games positive, break-even maker

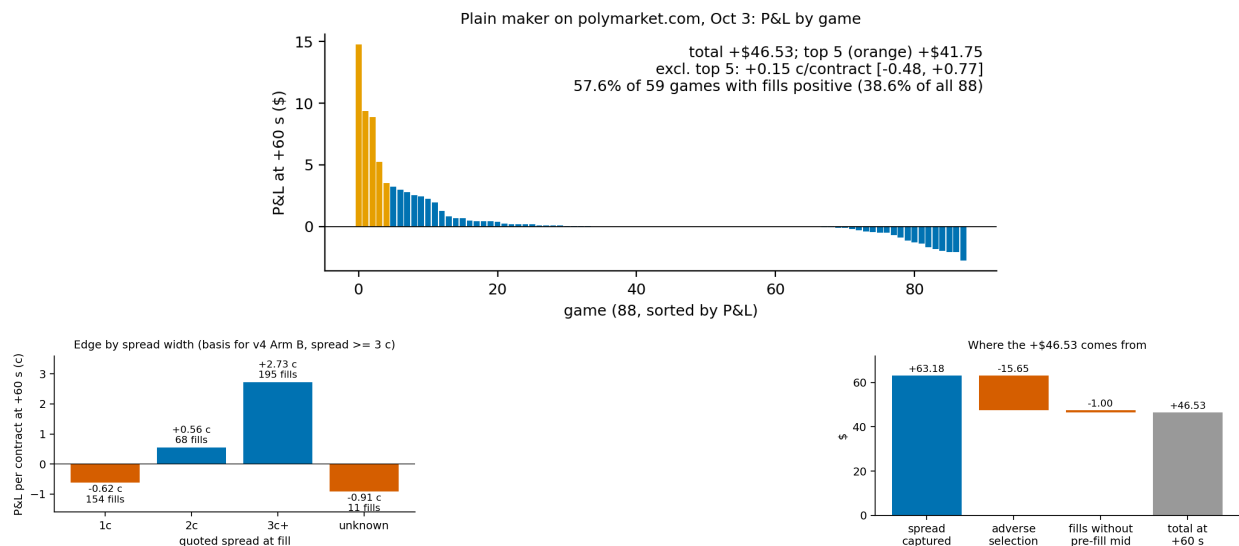


Figure 1: Origin of HYPOTHESIS_v4, plain polymarket.com maker on Oct 3. Top: P&L per game at +60 s, sorted; the top five games carry \$41.75 of \$46.53. Bottom left: P&L per contract by quoted spread (1 cent loses, 3 cents or more wins). Bottom right: spread captured versus adverse selection. Descriptive, one day, not pre-registered, simulated on recorded top of book, maker fee 0, queue position unknown. Full captions and the frozen v4 text: appendix, HYPOTHESIS_v4 section.

fee above 0.5 cents), so this is a lead worth testing live. The rule was written after these numbers were seen; read over all 88 games (38.6% positive) its game-share condition fails, which is why the live test decides. **Disclosure:** the origin run used Oct 3 and 4 games, inside the holdout period, with no committed authorization; it can motivate v4, never test it.

Why v4 is not checked on the earlier holdout. For Aug to Oct 2 we hold only polymarket.com taker trades at 1 s resolution, no order books, so the join and trade-through rule cannot be simulated faithfully. **Test plan.** Test set 1 had 0 eligible games (recordings ended before the late Oct 3 games finished). Test set 2 (confirmatory) is every college and NFL game from Oct 8 00:00 ET to Oct 13 00:00 ET, run once after the window closes. Decision, per venue and arm with Holm across venues: supported only if (1) the primary CI is above zero, (2) the point estimate excluding the top five games is above zero, and (3) the settlement P&L point estimate is above zero; otherwise not supported, reported either way.

2 Before the plot twist: the pre-registered v3 study (not v4)

Nothing in this part is v4; v4 has no result yet. This is the earlier pre-registered study that led to v4. Its strategies (the v3 strategies) did not survive their single holdout test, and 566 disclosed post-hoc trials found no other edge (lowest Reality Check p 0.42 over 363). The passive maker is the one idea that survived a reading, and it was frozen as v4 before any test data.

2.1 Economic hypothesis

A first CME versus Kalshi hypothesis was retired before any backtest (no CME NFL game in the holdout window). Before v4, we pre-registered two hypotheses and three Kalshi strategies (the v3 strategies A, A-maker and B; unrelated to v4's Arms A and B). **H1, lead-lag:** one venue's book reprices before another's over 1 to 15 s because participants and order handling

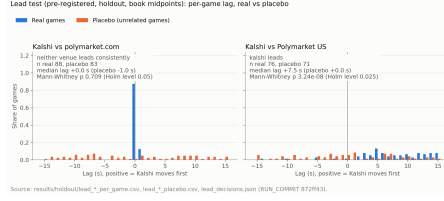


Figure 2: Pre-registered v3 lead test on the holdout (not v4): per-game lag of book midpoints, real games vs unrelated-game placebo pairs. Positive lag means Kalshi moves first.

differ (polymarket.com applies a taker delay and cancels resting sports orders at game start) [4, 9]. **H2, favorite-longshot bias:** bettors overpay for longshots [8, 7, 2]. Strategy **A** buys the pregame favorite as a taker, **A-maker** rests the same order, and **B** trades Kalshi toward polymarket.com when the two disagree for several seconds. Every amendment is a dated commit before the single holdout run (872ff43).

2.2 Data and universe

Kalshi: historical trades for every KXNFLGAME and KXNCAAFGAME market (training 2025-07-31 to 2026-01-25) and live websocket books for the holdout; Webull routes retail orders to it, and its fee is our primary cost. **polymarket.com:** trades for training, live books for the holdout (US persons cannot trade it; we only read its data). **Polymarket US:** polled quotes and time-and-sales, holdout only. ESPN kickoffs; two recorders never spliced within a game; every listed game kept, voids excluded and listed, ties settle at 0.5. An orientation gate checked that each game’s two Kalshi team markets sum to about one (n 670, 0 outside the band).

2.3 Methodology

No lookahead. A decision at t uses only rows at or before t (as-of joins); fills happen at t plus 1 s, never at t . **Sealed holdout.** Games with kickoff on or after 2026-08-01 were never loaded before the run; the run script refuses them otherwise. **Costs.** Webull \$0.02 per contract per fill (primary) and Kalshi direct fees, plus a half-spread; every result is also shown at costs $\times 2$; 10 contracts per order, never optimized. **Lead test.** Per game, the lag maximizing the cross-correlation of 1 s book-midpoint changes, compared with unrelated-game placebo pairs (Mann-Whitney, Holm), kickoff window cut. **Strategies.** A: 5 min before kickoff, buy the favorite if its price is at least $\theta \in \{0.70, 0.80, 0.90\}$, hold to settlement. B: enter when the venue gap $|d| \geq k$ for m seconds, exit when $|d| < \text{one cent}$ or after T seconds, $k \in \{3, 5\}$ cents, $m \in \{10, 30\}$ s, $T \in \{60, 300\}$ s. Selection on training only; confidence intervals are game-level bootstraps; Sharpe is mean over standard deviation of daily P&L (as a share of the capital base), $\times \sqrt{365}$, over every calendar day of the season with zero days kept. **Multiple testing.** Every setting tried, pre-registered or not, is a row in `experiments/variants.csv` (77 rows), and the deflated Sharpe ratio [1] is reported at the pre-registered and the final trial counts.

2.4 Results of the pre-registered study

Lead test. Kalshi vs polymarket.com: 88 qualifying games, median lag 0.0 s, 12.5% of games positive, placebo median -1.0 s, $p = 0.709$: “Neither venue leads consistently” (Figure 2). Kalshi vs Polymarket US: 76 games, median lag 7.5 s, 92.1% positive, $p = 3.24 \times 10^{-8}$ against a Holm level of 0.025: “Kalshi leads”. We report that outcome unchanged and explain below why we do not interpret it.

	A (taker, θ 0.80)		B		Combined	
	Train	Holdout	Train	Holdout	Train	Holdout
Trades	302	263	9,242	2,478	9,544	2,741
Net P&L (\$)	-21	-73	-4,496	-1,158	-4,517	-1,232
Ann. return (of capital)	-32%	-326%	n/a	n/a	-5,440%	-4,420%
Ann. volatility	53%	85%	n/a	n/a	628%	568%
Sharpe (ann.)	-0.60	-3.81	-8.59	-7.76	-8.66	-7.78
Sharpe, costs $\times 2$	-2.95	-6.03	-8.65	-7.84	-8.71	-7.89
Max drawdown (\$)	77	75	4,496	1,149	4,521	1,224
Turnover (contracts/day)	17	44	1,033	826	1,050	870
Daily skew	-1.12	-3.75	-3.07	-3.25	-3.03	-3.18
Worst month (\$)	-16	-30	-1,571	-661	-1,581	-678

Table 1: Pre-registered v3 strategies (not v4), net of Webull costs, 10 contracts per trade. Capital base = maximum capital committed at once on training (A \$137, combined \$170); B’s base is too small for return levels to mean anything, so B is reported per contract and by Sharpe. Training 2025-07-31 to 2026-01-25; holdout 2026-08-06 to 2026-10-04.

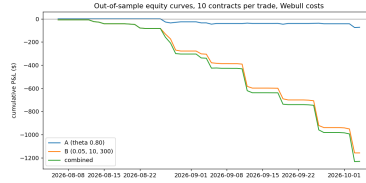


Figure 3: Holdout equity curve of the v3 strategies (not v4), 10 contracts per trade, Webull costs (training curve in the repository).

Traps. Trade-price lead-lag on training showed a Kalshi lead that unrelated-game placebos matched; in simulation with no true lead, thin trade prints fake leads while book midpoints do not [6, 5]. The v3 laggard strategy on Polymarket US (not v4) saw a lead only because the quote endpoint was cached (**max-age** 30 s): its fills on polled quotes look profitable (1.48 cents), but its fills confirmed by real trades lost -4.09 cents per contract (n 14). The v3 laggard against polymarket.com (not v4) loses -3.60 cents.

v3 Strategy A (favorite-longshot, taker; not v4). Selected $\theta = 0.80$ on training (return on capital -0.007 , 302 trades; flat plateau). v3 holdout: 263 trades in 768 games, return on capital -0.030 $[-0.069, 0.006]$; all three pre-registered fail conditions trigger. The bias is real in direction (the underdog at the same moment returns -0.463) but smaller than taker costs. **v3 A-maker (not v4):** 9 fills in 314 attempts, return on capital -0.150 : fills came when the market moved against the order (adverse selection), the risk v4 must overcome.

v3 Strategy B (cross-venue disagreement; not v4). Selected $k = 5$ cents, $m = 10$ s, $T = 300$ s on training (already negative). v3 holdout: 2,478 trades, -4.67 cents per contract $[-4.98, -4.36]$; gross edge is near zero and B pays a near-fixed round-trip cost on every trade, hence its very negative Sharpe.

v3 combined book (not v4). The combined v3 holdout P&L was $\$-1,232$ (Table 1, Figure 3). A’s training P&L has no factor exposure (alpha $t = -0.33$ on Mkt-RF, HML, UMD [3]). The deflated Sharpe ratio [1] is 1.27×10^{-9} for A and 1.18×10^{-34} combined at 22 pre-registered trials; counting all 566 disclosed trials can only lower it.

v3-era post-hoc work (after the holdout run; not v4). We tested 12 further ideas, a Polymarket US time-and-sales test and a book of every training-positive cell, each specified in a commit before its run and counted in the deflated Sharpe (appendix, Post-hoc work). None had an edge; the training-positive book returns -0.019 $[-0.060, 0.021]$ on the holdout. Descriptively, Kalshi beat ESPN’s win probability on calibration in all 8 variants. Walk-forward meta-models predict winners (AUC 0.77) yet lose (return on capital -0.121): predicting who wins is easy, predicting who beats the price is not. Later training-only searches added 485 trials; White’s Reality Check over the largest set (479, best $t = 2.50$) gives $p = 0.47$. **Trap 3:** a v3-era Kalshi versus polymarket.com arbitrage (not v4) showed 1.65 cents per contract only because it assumed the second leg always cleared; with one-legged fills counted it is -1.57 .

Robustness. Removing float residue levels from the recorded Kalshi books changes no midpoint and leaves both lead-test decisions unchanged (ec2c401).

3 Risk management

v4 quotes 10 contracts per side with an inventory cap of ± 50 contracts per game. Arm B quotes only while the spread is at least 3 cents, where the reward per fill was largest on the origin day. Fills count only on strict trade-through, a conservative rule. Settlement and regulatory risk apply (an early-settled and reversed Kalshi market; Nevada enforcement against Kalshi and Polymarket US). Before live trading we would add a daily loss limit, a per-venue exposure cap, a kill switch on feed outages over 60 s and fill reconciliation. All trading is simulated.

4 Liquidity and capacity

For a maker the binding constraint is queue position, not size. Queue position on the origin day is unknown, and strict trade-through makes the fill count a lower bound. Deep NFL books with 1 cent spreads may rarely reach a joining quote; thinner college books with wider spreads are where Arm B operates.

5 Limitations and next steps

The origin is one day of mostly college games, top of book only. The maker fee was 0 on Oct 3 and may change; test set 2 applies each venue’s published schedule at the time. polymarket.com is not available to US persons, so a tradable version needs Polymarket US, whose authenticated order book we have never recorded; the test set 2 recorder still has to be built and running before Oct 8.

6 Conclusion

HYPOTHESIS_v4 is a bet on the other side of the trade, in largely uncharted territory: that a patient maker is paid for supplying liquidity to recreational football flow. On one recorded day it captured 1.63 cents of spread per contract and lost 0.40 cents to adverse selection, net 1.09 $[0.17, 2.12]$ cents, concentrated in five games. High risk, high reward: a reason to test, not yet a result. v4 is frozen (94aa1a0) and will be tested once on Oct 8 to 12, and we will report the answer either way.

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